

Dunnett's Critical Value Table

One-Tailed Critical Values

Significance Levels: $\alpha = 0.05$ and $\alpha = 0.01$

k = 2 to 9 treatment groups | Error df = 2 to ∞

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When to use one-tailed Dunnett's test: Use the one-tailed table when you have a clear directional hypothesis — for example, all drug doses are expected to *increase* the response relative to placebo. This is common in dose-response studies, clinical trials with expected efficacy, and experiments where a treatment can only improve (or only worsen) an outcome. One-tailed critical values are smaller, giving more statistical power — but only when the direction is correct.

Caution: Choosing one-tailed based on observed data (data dredging) inflates the Type I error rate. The directional hypothesis must be specified *before* data collection.

Dunnett's One-Tailed Critical Values alpha = 0.05								
Use when all treatments are expected to differ from control in ONE direction only Yellow row = df = 20								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.02	2.43	2.68	2.85	2.99	3.10	3.19	3.27
3	1.88	2.22	2.43	2.58	2.70	2.80	2.89	2.96
4	1.82	2.13	2.33	2.47	2.59	2.68	2.76	2.83
5	1.78	2.07	2.26	2.40	2.51	2.60	2.68	2.75
6	1.76	2.04	2.23	2.36	2.47	2.56	2.63	2.70
7	1.74	2.02	2.20	2.33	2.44	2.52	2.60	2.66
8	1.73	2.01	2.18	2.31	2.41	2.50	2.57	2.64
9	1.72	1.99	2.17	2.30	2.40	2.48	2.55	2.62
10	1.71	1.98	2.16	2.28	2.38	2.47	2.54	2.60

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
11	1.71	1.97	2.15	2.27	2.37	2.45	2.53	2.59
12	1.70	1.97	2.14	2.27	2.36	2.45	2.52	2.58
13	1.70	1.96	2.14	2.26	2.36	2.44	2.51	2.57
14	1.70	1.96	2.13	2.25	2.35	2.43	2.50	2.57
15	1.70	1.96	2.13	2.25	2.35	2.43	2.50	2.56
16	1.69	1.95	2.12	2.24	2.34	2.42	2.49	2.56
17	1.69	1.95	2.12	2.24	2.34	2.42	2.49	2.55
18	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.55
19	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.54
20	1.69	1.95	2.12	2.24	2.33	2.41	2.48	2.54
24	1.68	1.94	2.11	2.22	2.32	2.40	2.47	2.53
30	1.68	1.93	2.10	2.21	2.31	2.38	2.45	2.51
40	1.68	1.92	2.09	2.20	2.30	2.37	2.44	2.50
60	1.67	1.91	2.08	2.19	2.28	2.36	2.43	2.49
120	1.66	1.91	2.07	2.18	2.27	2.35	2.41	2.47
∞	1.65	1.90	2.06	2.17	2.26	2.34	2.40	2.46

Source: Dunnett (1955), JASA 50:1096-1121. One-tailed values derived from the upper tail of the multivariate t-distribution.

Test Statistic $d = (\text{mean}_i - \text{mean}_0) / SE$	Error Degrees of Freedom $df_{\text{error}} = N - (k + 1)$
Standard Error $SE = \sqrt{MSE \times (1/n_i + 1/n_0)}$	Decision Rule Reject H_0 if $ d > d^*$

One-Tailed vs Two-Tailed: Example Comparison

Setting	k	df_error	alpha	Two-Tailed d*	One-Tailed d*	Difference
Common lab	3	20	0.05	2.54	2.19	-0.35 (one-tailed is less strict)
Pharma trial	4	30	0.05	2.61	2.27	-0.34
Strict testing	3	20	0.01	3.20	2.86	-0.34
Large study	5	60	0.05	2.57	2.28	-0.29

Note: One-tailed d* is consistently smaller than two-tailed d*, providing greater power when the direction of the effect is correctly anticipated.

Dunnett's One-Tailed Critical Values alpha = 0.01								
For strict directional testing — pharmaceutical, clinical trial, and regulatory use Yellow row = df = 20								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.92	3.34	3.61	3.81	3.96	4.09	4.20	4.29
3	2.54	2.87	3.08	3.24	3.37	3.48	3.57	3.65
4	2.39	2.68	2.87	3.01	3.13	3.22	3.31	3.38
5	2.30	2.57	2.74	2.88	2.99	3.08	3.16	3.23
6	2.24	2.50	2.67	2.80	2.90	2.99	3.07	3.14
7	2.20	2.46	2.62	2.75	2.85	2.93	3.01	3.07
8	2.17	2.43	2.59	2.71	2.81	2.89	2.97	3.03
9	2.15	2.40	2.56	2.68	2.78	2.86	2.93	3.00
10	2.13	2.38	2.53	2.65	2.75	2.83	2.90	2.97
11	2.12	2.37	2.52	2.64	2.73	2.82	2.88	2.95
12	2.11	2.36	2.51	2.63	2.72	2.80	2.87	2.94
13	2.10	2.35	2.50	2.62	2.71	2.79	2.86	2.92

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
14	2.10	2.34	2.49	2.61	2.70	2.78	2.85	2.91
15	2.09	2.34	2.49	2.60	2.70	2.77	2.84	2.91
16	2.09	2.33	2.48	2.60	2.69	2.77	2.84	2.90
17	2.08	2.33	2.48	2.59	2.68	2.76	2.83	2.89
18	2.08	2.33	2.47	2.59	2.68	2.76	2.82	2.89
19	2.08	2.32	2.47	2.58	2.67	2.75	2.82	2.88
20	2.07	2.32	2.47	2.58	2.67	2.75	2.81	2.88
24	2.06	2.31	2.45	2.56	2.65	2.73	2.79	2.86
30	2.05	2.29	2.43	2.54	2.63	2.71	2.77	2.83
40	2.04	2.28	2.42	2.53	2.61	2.69	2.75	2.81
60	2.02	2.26	2.40	2.50	2.59	2.66	2.73	2.78
120	2.01	2.25	2.38	2.49	2.57	2.65	2.71	2.77
∞	2.00	2.23	2.37	2.47	2.56	2.63	2.69	2.75

Source: Dunnett (1955), JASA 50:1096-1121. Validated against Kutner et al. (2005) and NIST/SEMATECH e-Handbook.

Document Information

Document Type	Statistical Reference Table — One-Tailed Dunnett's Table ($\alpha = 0.05 \text{ \& } 0.01$)
Author / Compiler	Statistics Fundamentals Editorial Team
Senior Reviewer	Minsa A. (Senior Statistics Editor)
Publisher	StatisticsFundamentals.com
Publication Date	May 07, 2026
Version	1.0 (May 2026)

Primary Source	Dunnett, C.W. (1955). JASA, 50(272), 1096-1121. DOI: 10.2307/2281208
Secondary Sources	Kutner et al. (2005). Applied Linear Statistical Models, 5th ed. McGraw-Hill; NIST/SEMATECH e-Handbook of Statistical Methods (itl.nist.gov)
Validation	Values cross-checked against Penn State STAT 503 course materials (online.stat.psu.edu/stat503) and R stats package DunnettTest()
License	Free for educational and non-commercial research use. Cite: StatisticsFundamentals.com/tables/dunnetts-table/
Keywords	Dunnett's table, Dunnett critical values, multiple comparisons, control group, post hoc test, ANOVA, family-wise error rate, FWER, one-way ANOVA, k treatment groups, error degree of freedom
URL	https://statisticsfundamentals.com/tables/dunnetts-table/

References & Sources

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